MIL-M-38313C (USAF)
15 September 1988
SUPERSEDING
MIL-M-38313B(USAF)
15 December 1980

MANUALS, SYSTEM, REAL PROPERTY INSTALLED EQUIPMENT, PREPARATION OF

This specification is approved for use by the Ballistic Missile Office, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for preparation of system manuals for operation and maintenance of Real Property Installed Equipment (RPIE).

1.1.1 This specification also provides requirements for preparation of a combination system-equipment manual.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Ballistic Missile Office/AWD, Norton AFB, CA 92409 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC F4531 AREA TMSS

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2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and Standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein (see 2.4.1).

SPECIFICATIONS

Military

MIL-M-38312	Manuals, Equipment, Real Property Installed Equipment, Preparation of
MIL-M-38784	Manuals, Technical: General Style and Format Requirements
MIL-P-38790	Printing Production of Technical Manuals, General Requirement for (Photolithographic Negatives, Reproduction Assembly Sheets, Printing)

STANDARDS

Military

DOD-STD-863	Wiring Data and System Schematic Diagrams, Preparation of
MIL-STD-12	Abbreviations for Use on Drawings, and in Technical Documents
MIL-STD-17-1	Mechanical Symbols
MIL-STD-1247	Marking Functions and Hazard Designations of Hose, Pipe, and Tube Lines for Aircraft Missile and Space Systems
MIL-STD-1752	Reading Level Requirements for Preparation of Technical Orders

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal apply (see 2.4.2 and 2.4.3).

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ANSI Y14.15-1966 (Rev. 8/73)	Electrical and Electronic Diagrams
ANSI Y32.9-1972	Graphic Symbols for Electrical Wiring and Layout Diagrams Used in Architecture and Building Construction
ANSI/IEEE 260-1978	Letter Symbols for Units Used in Science and Technology (formerly ANSI Y32.14-1973)
IEEE STD 91-1973	Graphic Symbols for Logistic Diagrams (Two-State Devices) (same as ANSI Y32.14-1973)
IEEE STD 100-1975	Reference Designations for Electrical and Electronic Parts and Equipment (same as ANSI Y32.16-1975)
IEEE STD 315-1975	Graphic Symbols for Electrical and Electronic Diagrams (Including Reference Designation Class Designation Letters) (same as ANSI Y32.2-1975)

2.3 Order of Precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

2.4 Source of Documents.

2.4.1 Government Specifications and Standards. Copies of the referenced military specifications and standards are available from the Department of Defense Single Stock Point, Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. For specific acquisition functions, these documents should be obtained from the contracting activity or as directed by the contracting activity.

2.4.2 Other Government Documents. Copies of other government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity. The documents listed may be obtained as follows:

a. Copies of listed documents are available from the Department of Defense Single Stock Point, Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

b. Copies of the GPO Style Manual should be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

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c. Copies of the DOD Thesaurus of Engineering and Scientific Terms and The DDC Retrieval and Indexing Terminology may be purchased from the National Technical Information Service, Springfield, VA 22161, or (for DOD activities and DOD contractors) from the Reference Services Branch, Defense Technical Information Center, Cameron Station, Alexandria, VA 22304-6145.

2.4.3 Sources for Nongovernment Publications. Nongovernment documents are generally available for reference from libraries and technical groups. The documents listed may be obtained from the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.

3. REQUIREMENTS

3.1 System Manual. RPIE system manuals shall be prepared for each RPIE functional system or homogeneous grouping of functional systems as identified by the contractor and approved by the contracting activity. The manual(s) shall provide instructions for operation and maintenance of the complete system. The functional system breakout and the number of manuals required shall be determined by the contracting activity. (See 6.1)

3.1.1 System-Equipment Manual. Manuals may also be prepared in a combination system-equipment format when such action is determined to better serve the needs of the using command and approved by the procuring activity. (See 6.1).

3.2 Manner of Preparation. Unless otherwise specified herein, each RPIE system manual shall be prepared in general accordance with MIL-M-38784.

3.2.1 System Equipment Manual. When a combination system-equipment manual is prepared, the system portion and front matter shall be prepared as specified herein. The equipment portion shall be prepared in accordance with MIL-M-38312.

3.2.2 Nomenclature Consistency. Nomenclature shall be consistent throughout directly related RPIE system manuals.

3.2.3 Abbreviations. Use of abbreviations shall be limited to those specified in MIL-STD-12 and the glossary (see 3.15).

3.2.4 References. Whenever the functional responsibility within a given RPIE system relies on a functional responsibility of another system or other equipment, such a correlation or continuation shall be noted and reference made to the applicable manual. This reference applies to text, tables, illustrations, and drawings.

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3.2.5 Page Numbers and Titles. Pages, paragraphs, and illustrations shall be numbered separately and consecutively within each section by arabic numerals. The forming of arabic numerals shall be in accordance with MIL-M-38784; however, in instances where more than one RPIE system is presented in the same manual, a part or chapter designation shall also appear on the upper outside corner of the page. Chapters (or parts), sections, paragraphs, and illustrations shall have a brief title describing content or action portrayed.

3.2.6 Page Layout.

3.2.6.1 Copy Arrangement. Arrangement of copy shall be in accordance with MIL-M-38784.

3.2.6.2 Space Conservation. Layout shall conserve space without lessening usability or clarity of material. Blank spaces shall be avoided whenever possible.

3.2.6.3 Title and Headings. Section headings shall be centered horizontally on the page.

3.2.7 Number Identification of Manual. The activity preparing the manuals shall place an identifying number on the manual. This number will be provided by the procuring activity.

3.2.8 Depth of Coverage. The manuals shall cover all of the equipment which comprises a complete functional system. However, the text shall not duplicate information contained in the individual equipment manuals unless essential to describe the system operation and maintenance activities.

3.2.9 Style of Instructions. The manuals shall portray an understanding of how the systems operate in the operating environment and shall provide sufficient procedures for operation and maintenance. Presentation shall conform with MIL-M-38784.

3.2.10 Style of Writing. The instructions shall be concise, specific, clearly worded, and illustrated. Descriptive material shall be held to a minimum. Details of procedures which are obvious or generally known, such as removing screws, general cleaning instructions, or normal shop practices, need not be given.

3.2.10.1 Charts, Checklists and Diagrams. Use of charts, checklists, and diagrams presented as figures in abbreviated columnar data format shall be employed, whenever practical, in lieu of written text (see Figure 1).

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STEP	PROCEDURE	NORMAL INDICATION	CORRECTIVE ACTION
	CAUTION Ensure that all conditions listed in prestarting procedures exist before starting the system (Fig. 2-20). Failure to comply could result in equipment damage. **NOTE** Steps 1 through 7 are performed at BM panel (PNL-3D068-0) on silo level 6.		
1.	Set SILO AIR COND SYS START ON-OFF switch to ON.		
2.	Check SILO AIR COND SYS START indicator.	Lighted.	Check indicator lamp and fuze, or see 1, fig. 2-29.
3.	Check EF-102 indicator.	Lighted.	Check indicator lamp or see 2, fig. 2-29.
4.	Check AC-102 indicator.	Lighted.	Check indicator lamp or see 4, fig. 2-29.
5.	Check P-110 indicator.	Lighted when TC-100 branch air pressure is less than 7.5 psi.	Check indicator lamp or see 5, fig. 2-29.
6.	*Deleted.*		
7.	Check dehumidifier D-101 OPERATING LIGHT	Lighted.	Check indicator lamp or see 21, fig. 2-29.

Figure 2-21. Launch Duct Air Conditioning System Starting Procedures

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FIGURE 1. Typical Columnar Format

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STEP	PROCEDURE	NORMAL INDICATION	ABNORMAL INDICATION	CORRECTIVE ACTION
	NOTE All indicator lights reflected in the NORMAL INDICATION column are located on the MPP unless otherwise indicated.			
1.	Notify MCCC of task to be performed.			
2.	Set STANDBY POWER main circuit breaker MCB in Generator Control Panel to OFF.	ENG OVERSPEED . ON STBY PWR FAIL . ON	ENG OVERSPEED . OFF STBY PWR FAIL . OFF	Proceed to step 3. Replace MCB circuit breaker and proceed to step 3.
3.	Set STANDBY POWER main circuit breaker MCB in Generator Control Panel to ON.	ENG OVERSPEED . OFF STBY PWR FAIL . OFF	ENG OVERSPEED . ON STBY PWR FAIL . ON	Proceed to step 4. Replace MCB circuit breaker and proceed to step 4.
	NOTE If Model M-1 digital frequency meter is used, refer to paragraph 1-126.			
4.	Connect portable frequency meter to 120-vac receptacle on Generator Control Panel; then, proceed to step 5.			
	NOTE DEU coolant temperature must be stabilized prior to checkout or adjustment of overspeed governor.			
5.	Perform steps 1 thru 9 of LCEB and LFSB Standby Power System Operation in Manual Mode procedures (fig. 2-2); then, proceed to step 6.			

Figure 3-12. LCEB and LFSB Diesel-Electric Unit Overspeed Safety Circuit Checkout and Adjustment (Sheet 1 of 5)

3-15

FIGURE 2. Columnar Format (Arabic)

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1. Commercial power failure.

COML PWR FAIL . . . . . ON (Not blinking)
FAIL TO TRANS COML PWR . . . . . OFF

PROCEDURE	NORMAL INDICATION	ABNORMAL INDICATION	CORRECTIVE ACTION
a. Verify commercial power main circuit breaker is set to ON.	ON.		Proceed to step b.
		OFF.	Set main circuit breaker to ON.
WARNING Hazardous voltages are present in Power Transfer Switch Cabinet. While performing checks and/or maintenance, lock door(s) in open position and observe standard safety precautions.			
b. Check and record phase to neutral voltages (Phase A to neutral, Phase B to neutral, and Phase C to neutral).	120 (± 8) vac.		Proceed to step c.
		Out of tolerance.	Notify power company.
c. Computer and record average voltage by adding voltages recorded in step b., then divide total by 3.	All voltage readings recorded in step b. are within 10 volts of average.		Proceed to step d.
		All voltage readings recorded in step b. are not within 10 volts of average.	Notify power company.
d. Check phase to neutral voltages at TB1 as follows: Phase A, TB1-7 Phase B, TB1-9 Phase C, TB1-11 Neutral, TB1-13	120 (± 8) vac.		Replace MPP.
		Out of tolerance.	Proceed to step e.

Figure 3-6. LCF Power Generation and Distribution System Troubleshooting (Sheet 2 of 24)

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FIGURE 3. Columnar Format (Numeric and Alpha)

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3.2.10.2 Steps. Steps or procedures contained within framed figures shall be listed in arabic numbers (see Figure 2) except in unique cases where trouble analyses require troubles to be listed in numerics. In such cases, steps or procedures should be presented in alphas (see Figure 3).

3.2.10.3 Columnar Format. Requirements such as checkout, adjustment, and troubleshooting will normally consist of five columns: step, procedure, normal indication, abnormal indication, and corrective action. Circumstances may dictate departures from this general format.

3.2.11 Art. Except as otherwise specified, art for system manuals shall be prepared as specified in MIL-M-38784, to include titles and numbering. No illustration shall exceed 11 x 38 inches paper size, and every effort will be exerted to keep the size no larger than 11 x 23 inches. Line drawings shall be used in lieu of photographs (halftones). Use of isometric and perspective-type illustrations is permitted in instances where such illustrations do not detract or lessen the graphic presentation. Multiple-sheet illustrations shall be numbered in accordance with MIL-M-38784.

3.2.11.1 Graphic Symbols. Graphic symbols used on illustrations shall be per "STANDARDS" and "Other Publications" list in "APPLICABLE DOCUMENTS" (see 2.1 and 2.2). Exceptions shall be approved by the procuring activity in instances where nonstandard symbols more clearly define and support the functional tasks of the system involved. All nonstandard symbols shall be defined in the symbol chart glossary (see 3.15).

3.2.11.2 Reference Designation. Reference designation shall be in accordance with IEEE STD 100-1975. Exceptions shall be approved by the procuring activity in instances where nonstandard reference designators more clearly define and support the task to be performed. All nonstandard reference designators shall be defined in a legend on the illustration or in the text supported by the illustration.

3.2.11.3 Existing Illustrations. Existing illustrations and engineering drawings shall be used where they satisfy the requirements herein. Specifically, wiring data and schematic diagrams prepared to DOD-STD-863 shall be used without redrawing.

3.2.12 Relaxed Format and Reproduction Requirements. Relaxed format and reproduction methods shall be permitted for system manuals in the interest of expeditious availability and economy. Areas which may be relaxed are as follows:

a. Use of single column format (type continuous across page).

b. Use of standard typewriter to prepare reproduction copy.

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c. Use of free hand lettering on illustrations.

d. Use of office-type reproduction equipment.

e. Use of nonuniform lettering size on final copy. (Not smaller than 6-point type.)

f. Use of commercial illustrations and drawings to maximum possible extent.

g. Printing on one side of page only.

3.2.13 Reading Grade Level (RGL). The manuals shall be written at an RGL commensurate with the capability of the target audience. MIL-STD-1752 shall be used as a guide to determine required RGLs. The methodology of MIL-M-38784 shall be used to determine the RGL.

3.3 Arrangement of Sections. Each RPIE system manual shall be sectionalized as follows:

Title Page
"A" Page
Table of Contents
List of Illustrations
Introduction
Safety Precautions

Section I	Description of System
Section II	Operating Instructions
Section III	System Maintenance
Section IV	Wiring Diagrams

Glossary

3.3.1 Alternate Format. As directed by the procuring activity and in those instances where system manuals will be used exclusively by Strategic Missile Wing personnel, or there is a need to parallel the format of a related technical order, an alternate format will be developed to include the following six sections:

a. Description

b. Special Tools and Test Equipment

c. Operation, Servicing and Checkout

d. Trouble Analysis

e. Launch Facility and Launch Control Facility Maintenance

f. Diagrams

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3.3.2 Sectionalization. If the complexity or inherent design of the system does not lend itself to the sectionalization indicated, separate sections shall be established for subsystem increments or major components of the functional system.

3.3.3 Parts and Chapters. In those instances where it is desirable to include all major support systems of a facility within one manual, the format will reflect parts and chapters as well as sections. For example:

Part II Air Conditioning, Heating and Ventilating Systems
Chapter 1 Environmental Control Systems
Section III System Maintenance

3.3.4 System-Equipment Manual. When the combination system-equipment manual is prepared, it will include the following four sections:

a. Description

b. Operation

c. Maintenance

d. Manufacturer's Operation and Maintenance Data

3.4 Title Page. The title page shall be prepared under a relaxed format. The most economical, best available method shall be used (see Figure 4).

3.4.1 System-Equipment Manual. The title page should be prepared under a relaxed format (see Figure 5).

3.4.2 Identifying Number. The identifying number shall be positioned in the upper right hand corner (see 3.2.7). Where the manual contains multiple systems, the Part and Chapter will also be placed in the upper right hand corner. For example:

SAC CEM 21-SM80B-2-28-4
Part III Chapter 2

3.4.3 Title. The title shall designate the type of manual, identify project and location applicability, and specify the facility, system, or item of equipment covered in the manual.

3.4.4 Authority Notice. The title page shall contain an authority notice that will identify the agency having management responsibility for the operational system.

3.4.5 Contract Number. The contract number shall be shown to identify the contract under which the manual was prepared.

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SAC CEM 35R-1-692-1

SYSTEM MANUAL

REAL PROPERTY INSTALLED EQUIPMENT

MISSILE WEAPON SYSTEM

USAF MODEL

LGM30

VANDENBERG AIR FORCE BASE

SUPPORT FACILITIES AS&I BUILDING

PUBLISHED UNDER AUTHORITY OF THE COMMANDER IN CHIEF, SAC

15 JANUARY 1974

FIGURE 4. Typical Title Page - System Manual

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SAC CEM 21-SM68B-2-26-2-1

SYSTEM AND EQUIPMENT MANUAL

REAL PROPERTY INSTALLED EQUIPMENT

MISSILE WEAPON SYSTEM

USAF MODEL

LGM25C

DAVIS-MONTHAN AIR FORCE BASE

PROPELLANT TRANSPORTER SHELTER

PUBLISHED UNDER AUTHORITY OF THE COMMANDER IN CHIEF, SAC

17 MAY 1980
Change 3 - 1 March 1983

FIGURE 5. Typical Title Page - System and Equipment Manual

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3.4.6 Publication Date. The publication date shall be the date the last material to be included was received (copy freeze date). The date shall be shown in the lower right corner.

3.4.7 Changes. Title pages for manual changes shall be the same as for basic manuals except that a change notice strip-in, replacement notice, and the date of change shall be added. These shall be prepared in general as outlined in MIL-M-38784 (see Figure 6).

3.5 "A" Page. A "List of Effective Pages" or "A" page shall immediately follow the title page. This page shall be prepared in accordance with MIL-M-38784.

3.6 Table of Contents. The table of contents shall list numbers and titles of sections, parts, chapters (when used), and main paragraph with their initial page numbers.

3.7 List of Illustrations. The list of illustrations shall list the number, title, and page number of each illustration.

3.8 List of Tables. The list of tables, if used, shall list the number, title, and page number of each table.

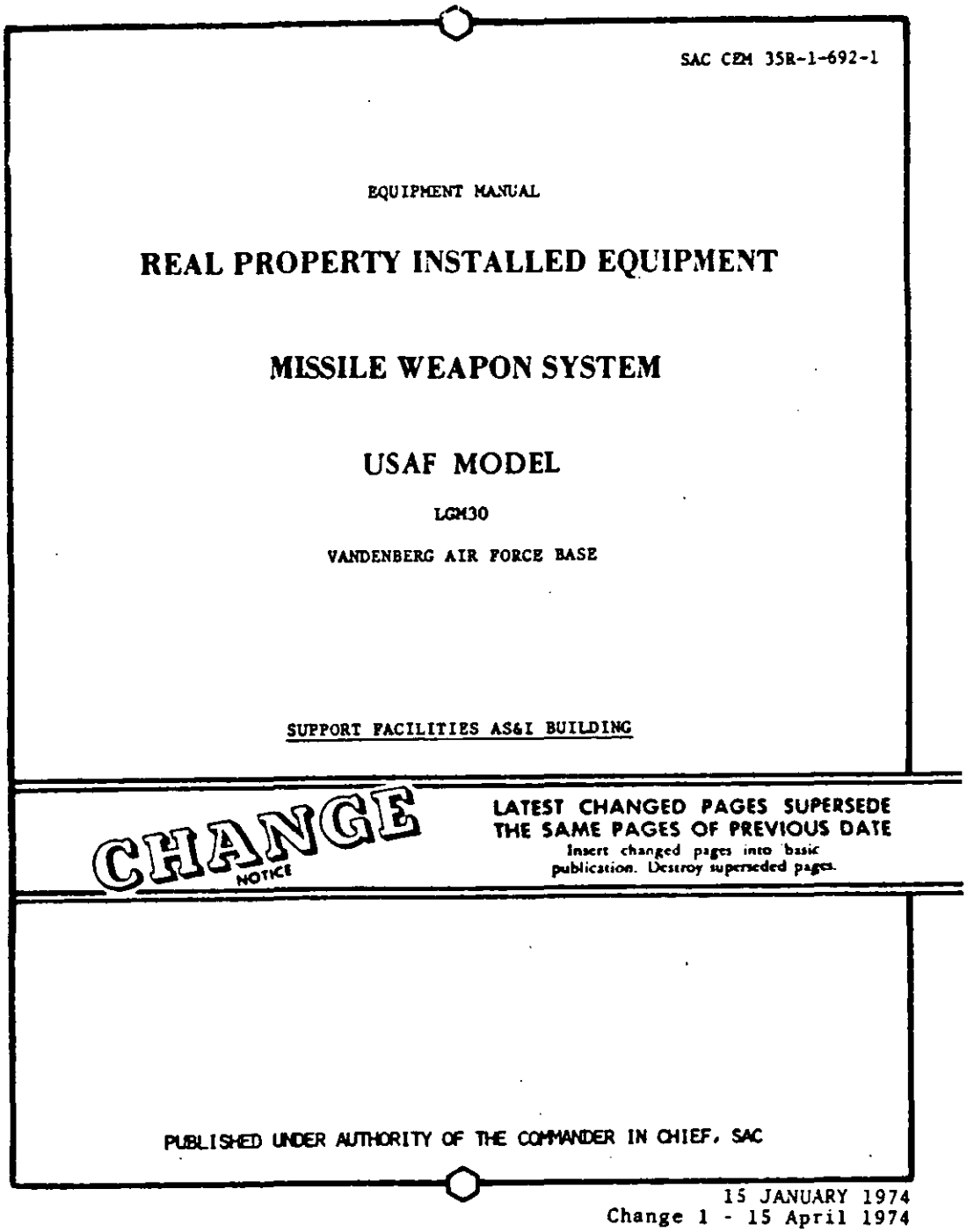
3.9 Introduction. The introduction to the manual shall contain a brief explanation of the functional RPIE system(s) covered by the manual. The introduction shall be concise and shall not contain a detailed description of equipment. The primary purpose of the introduction is to provide a ready orientation as to the use and purpose of the manual and its relationship to other RPIE system and equipment manuals (if applicable). This relationship shall be shown by adequate cross reference that will allow the user to quickly identify the referenced manual.

3.10 Safety Precautions. This information shall describe all major hazards to personnel and equipment safety that are peculiar to the equipment or jobs covered in the manual. The information should be in tabular form wherever practical; standard or common safety precautions and standard shop practices shall not be included. This shall in no way preclude the requirements for specific hazard information, caution, or warning notes to be interspersed at appropriate points throughout other sections of the manual.

3.11 Section I, Description of the System. This portion of the manual shall provide a physical and functional description of the system in its operating environment but shall not describe the system components. This section shall not contain procedures for performing operation and maintenance.

3.11.1 General Description. This portion shall describe the physical features of the system and its general arrangement. It shall also provide a brief description of the major components of the system, their location, and

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EQUIPMENT MANUAL

REAL PROPERTY INSTALLED EQUIPMENT

MISSILE WEAPON SYSTEM

USAF MODEL

LGM30

VANDENBERG AIR FORCE BASE

SUPPORT FACILITIES AS&I BUILDING

CHANGE NOTICE

LATEST CHANGED PAGES SUPERSEDE THE SAME PAGES OF PREVIOUS DATE

Insert changed pages into basic publication. Destroy superseded pages.

PUBLISHED UNDER AUTHORITY OF THE COMMANDER IN CHIEF, SAC

15 JANUARY 1974
Change 1 - 15 April 1974

FIGURE 6. Typical Title Page with Change Notice

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relationship of the complete system. The description of the major components shall be presented in tabular form to the maximum extent feasible. Appropriate illustrations, diagrams, and pictorial views shall be used in lieu of text whenever appropriate. The extent, content, and limitations of maintenance covered by the manual shall be explained in this section. This shall include special circumstances and environmental consideration, if applicable. If the basic system is comprised of several independent subsystems which are integrated into a major functional system, the subsystems may be covered separately in the same manual. However, in these cases, the relationship of the subsystem to the major system shall be explained.

3.11.2 Tools and Equipment Listing. A tabular listing shall be provided, with appropriate illustrations, of the special tools, equipment, and test equipment applicable to the procedures contained in the manual.

3.11.3 Description of the System Functions. This portion shall explain how the system functions as an integral unit. The degree of detail will depend to a large extent upon the complexity of the system. In some cases, a detailed description of the function may not be warranted. In such cases, the description of system functions may be combined with the general description portion of the manual. Flow diagrams, schematic diagrams, and functional loops necessary for understanding system functions and performing trouble analysis shall normally be included in this section. Systems shall be depicted in their normal operating mode. Appropriate text shall be used to explain the logic of actions illustrated in the flow diagrams. The manual shall not include basic textbook type theory of report basic information contained in equipment manuals. Unusual theoretical information may be included when required to fully explain the system functions.

3.12 Section II, Operating Instructions. This section shall contain only the essential information required for normal operation of the system. This shall include instructions necessary for setting up or preparing the equipment for use, warm-up procedures (if applicable), starting procedures, verification of normal operation, shutdown procedures, emergency procedures, and post-shutdown requirements. If operating instructions required preparation of detailed step-by-step procedures, or specific sequence of accomplishment, this information shall be presented in tabular, checklist form to the maximum extent possible. Whenever procedural difficulty or equipment complexity warrants, or wherever otherwise appropriate, a "show and tell" illustration-text integrated format should be used (see Figure 7).

3.12.1 Emergency Procedures. Emergency procedures shall consist of actions to be taken in the event of malfunction. Safe shutdown, backout, and methods of return to safe condition shall be included. Generally, emergency procedures shall be written only when damage to equipment or injury to personnel is possible.

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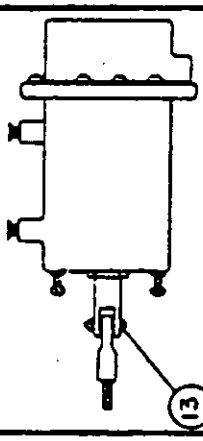
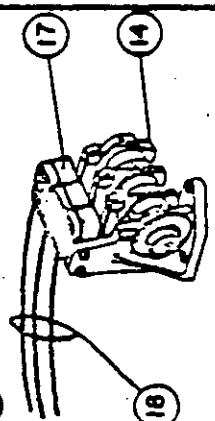
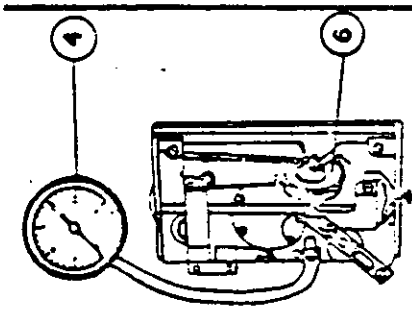
STEP	PROCEDURE	ILLUSTRATION	NORMAL INDICATION	CORRECTIVE ACTION
D.	(Continued)			
	15. Reconnect piston operator linkage by reinstalling screw/cotter pin (13).			
	16. Rotate TC-121 setpoint dial (6), as required, until test gage (4) indicates 13 psi.		All micro switches (17) are actuated.	Repeat steps 10 thru 16.
	17. Set HC-102 or HC-107 circuit breakers on MCC-2 panels (PNL-1C002-5, PNL-1C002-6, and PNL-1C002-7) to ON and remove DANGER tags.	**WARNING** Hazardous voltage present at step controller.		
	18. Using a clamp-on ammeter, measure and note amperage present on load side lead (18) of each micro switch (17).		Ammeter indicates approximately same amperage present on load side lead (18) of each micro switch (17). A variation of 20% or more from average indicates heater or circuit failure.	Check HC-102 or HC-107 circuit breakers, micro switches (17), or heater element. Adjust, repair, or replace defective components.
	19. Rotate TC-121 setpoint dial (6), as required, until test gage (4) indicates 5 psi.		All cams (14) release micro switches (17).	

Figure 1-15. Control Center Air Conditioning, Heating, and Ventilating System Adjustment Procedures (Sheet 16 of 27)

FIGURE 7. Illustration-Text Integrated Format

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3.13 Section III, System Maintenance. This section shall contain the checkout, troubleshooting, servicing, lubrication, remove and replace, and repair procedures performed on a system.

3.13.1 Checkout Procedures. This portion shall provide step-by-step checkout procedures required to verify the satisfactory operation of the system or major subsystems as applicable. This shall include any basic conditions or modes that must be achieved prior to checkout. If the checkout requires detailed step-by-step procedures, such procedures shall normally be presented in tabular form. These instructions shall clearly indicate why the checkout is performed and what conditions are to be sought. When malfunctions occur at this level, reference to the appropriate troubleshooting procedure shall be made.

3.13.2 Troubleshooting Procedures. These procedures, provided in columnar form, shall serve as a guide in isolating the faulty component. They shall be sequenced in logical progression from malfunction indication to location of the faulty element, and shall indicate the corrective action required. Any special connections or test equipment required for the troubleshooting shall be indicated at the appropriate point.

3.13.3 Servicing. Servicing requirements shall include cleaning, lubrication, replenishment of fuel, addition of refrigerant gas to the air conditioning system, replacement procedures, and other preventive maintenance procedures that apply to the particular equipment.

3.13.4 Remove and Replace Procedures. These procedures shall be provided only when they are not provided in the applicable equipment manuals. The instructions shall provide step-by-step procedures for removal and replacement of items which are subject to frequent replacement. These instructions should provide for judicious combination of text and illustrations. Obvious detail steps should be omitted.

3.13.5 Repair Procedures. Only those repair procedures which must be performed in-place or on a system-basis shall be provided. These instructions shall provide the necessary information to bring the equipment up to the required serviceable standard when it becomes unserviceable. If checkout is required to verify satisfactory operation of the equipment, reference to the appropriate section of the manual shall be made. Reference to the equipment manuals shall be used wherever possible.

3.14 Section IV, Wiring Diagrams. This section shall include wiring diagrams and associated data necessary for maintenance. The wiring diagrams shall be of the detailed schematic type, prepared separately for each particular circuit and equipment installed, to show all wiring connections and items of electrical equipment. Complete internal wiring of equipment shall not be shown. Existing wiring data and schematic diagrams prepared in accordance with DOD-STD-863 shall be used directly without redrawing. A symbol chart shall be provided to explain graphic symbols that appear on the

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diagram but are not covered in the standards listed in Section 2. A gas, fluid, and electrical conduit line identification chart shall be provided to explain identifications of lines that appear on the diagrams but are not covered in MIL-STD-1247.

3.15 Glossary. The glossary shall define terms used in the text that require definition because of their unique application to the weapon system and are not covered in military standards. It shall also include definitions of all nonstandard abbreviations, symbols, and line identifications used in the manual. The glossary page numbers shall be identified by a "G" followed by an arabic numeral.

3.16 Drilling and Binding. Drilling shall be in accordance with MIL-P-38790. (Special drilling for three-post or three-ring binder.) Manuals shall be delivered unbound, but contents shall be assembled and identified as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for Compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.2 Acceptance. Material furnished in accordance with this specification shall be accepted in accordance with the quality assurance provisions of MIL-M-38784 unless otherwise specified in the contract.

5. PACKAGING

5.1 Packaging, Packing, and Marking. Unless otherwise stated in the contract, packaging, packing and marking for shipment shall be in accordance with MIL-M-38784 and MIL-P-38790.

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6. NOTES

6.1 Data Requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements.

Para. No.	Data Requirements Title	DID No.	Option
3.1, 3.1.1	Real Property Installed Equipment (RPIE) Manuals	DI-TMSS-80677	None

(Copies of data item descriptions required by the contractor in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting activity.)

6.2 Definitions.

6.2.1 Real Property. The fixed capital assets of the government consisting of lands and interests therein, such as leases, permits, easements, licenses or rights-of-way, government-owned or leased grounds, structural facilities, and facility plants and systems. It includes those items of nonexpendable equipment and property which are attached to or installed in real property.

6.2.2 Real Property Installed Equipment (RPIE). Items attached to or installed in real property. RPIE is normally programmed, procured, funded, and installed through the Military Construction Program (MCP).

6.2.3 Equipment Manual. Technical document consisting of an organized assembly of maintenance and operating manuals, bulletins, brochures, and other data normally furnished by the manufacturer with his equipment, and supplemented as required.

6.3 Ordering Data. The contracting activity will specify whether:

a. A manual is to be used exclusively by Strategic Missile Wing personnel.

b. A manual is to parallel the format of a related technical order.

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6.4 Subject Term (Key Word) Listing.

Real Property Installed Equipment
Strategic Air Command Civil Engineering Manual
System-Equipment Manual
System Manual

6.5 Changes From Previous Issue. Vertical lines or asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CUSTODIAN:

Air Force - 14

USER ACTIVITIES:

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PREPARING ACTIVITY:

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(Project TMSS-F504)

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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